

Assignment No	15
Title	Revision Batch 1
Objective	Q1 (SQL – Data Partitioning) Q2 (R – Basics & Objects) Q3 (Pentaho – Table to Table ETL)
Roll No	MCA2511

Q1 (SQL – Data Partitioning)

Create a table Sales with appropriate attributes and implement Range partitioning on the sale_date column. Insert sample records and display data from each partition.

Solution:

```
SQL*Plus: Release 12.1.0.2.0 Production on Sat Dec 20 12:49:50 2025

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Enter user-name: system
Enter password:
Last Successful login time: Sat Dec 20 2025 12:48:25 +05:30

Connected to:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 - 64bit Production
With the Partitioning, OLAP, Advanced Analytics and Real Application Testing options

SQL> drop table sales1;

Table dropped.

SQL> create table sales1 (
  2   sale_id number(5),
  3   product_id number(5),
  4   customer_id number(5),
  5   time date,
  6   qty number(3),
  7   amount number(10)
  8 )
  9 partition by range (time) (
10   partition sale_q1 values less than (to_date('31-MAR-2005','dd-MON-yyyy')),
11   partition sale_q2 values less than (to_date('30-JUN-2005','dd-MON-yyyy')),
12   partition sale_q3 values less than (to_date('30-SEP-2005','dd-MON-yyyy')),
13   partition sale_q4 values less than (to_date('31-DEC-2005','dd-MON-yyyy'))
14 );

Table created.
```

```
SQL> insert into sales1 values (101,23,134,'15-Jan-2005',12,54561);
1 row created.

SQL> insert into sales1 values (102,52,789,'30-Mar-2005',545,52312);
1 row created.

SQL> insert into sales1 values (103,53,415,'11-Apr-2005',42,7683);
1 row created.

SQL> insert into sales1 values (104,45,365,'9-Jun-2005',54,2389);
1 row created.

SQL> insert into sales1 values (105,72,457,'2-Jul-2005',23,563912);
1 row created.

SQL> insert into sales1 values (106,8,865,'25-Sep-2005',21,635612);
1 row created.

SQL> insert into sales1 values (107,87,544,'12-Oct-2005',20,54662);
1 row created.

SQL> insert into sales1 values (108,75,341,'5-Dec-2005',45,54623);
1 row created.
```

```
SQL> select table_name, partition_name, high_value,num_rows FROM
2 user_tab_partitions where table_name = 'SALES1';
```

TABLE_NAME	PARTITION_NAME	HIGH_VALUE	NUM_ROWS
SALES1	SALE_Q1	TO_DATE(' 2005-03-31 00:00:00', 'SYYYY-MM-DD HH24:MI:SS', 'NLS_CALENDAR=GREGORIA	
SALES1	SALE_Q2	TO_DATE(' 2005-06-30 00:00:00', 'SYYYY-MM-DD HH24:MI:SS', 'NLS_CALENDAR=GREGORIA	
SALES1	SALE_Q3	TO_DATE(' 2005-09-30 00:00:00', 'SYYYY-MM-DD HH24:MI:SS', 'NLS_CALENDAR=GREGORIA	
SALES1	SALE_Q4	TO_DATE(' 2005-12-31 00:00:00', 'SYYYY-MM-DD HH24:MI:SS', 'NLS_CALENDAR=GREGORIA	

```
SQL> select * from sales1 partition(sale_q1);
```

SALE_ID	PRODUCT_ID	CUSTOMER_ID	TIME	QTY	AMOUNT
101	23	134	15-JAN-05	12	54561
102	52	789	30-MAR-05	545	52312

```
SQL> select * from sales1 partition(sale_q2);
```

SALE_ID	PRODUCT_ID	CUSTOMER_ID	TIME	QTY	AMOUNT
103	53	415	11-APR-05	42	7683
104	45	365	09-JUN-05	54	2389

```
SQL> select * from sales1 partition(sale_q3);
```

SALE_ID	PRODUCT_ID	CUSTOMER_ID	TIME	QTY	AMOUNT
105	72	457	02-JUL-05	23	563912
106	8	865	25-SEP-05	21	635612

```
SQL> select * from sales1 partition(sale_q4);
```

SALE_ID	PRODUCT_ID	CUSTOMER_ID	TIME	QTY	AMOUNT
107	87	544	12-OCT-05	20	54662
108	75	341	05-DEC-05	45	54623

QUERY:

```
create table sales1 (
  sale_id number(5),
  product_id number(5),
  customer_id number(5),
  time date,
  qty number(3),
  amount number(10)
)
partition by range (time) (
  partition sale_q1 values less than (to_date('31-MAR-2005','dd-MON-yyyy')),
  partition sale_q2 values less than (to_date('30-JUN-2005','dd-MON-yyyy')),
  partition sale_q3 values less than (to_date('30-SEP-2005','dd-MON-yyyy')),
  partition sale_q4 values less than (to_date('31-DEC-2005','dd-MON-yyyy'))
);
```

```
insert into sales1 values (101,23,134,'15-Jan-2005',12,54561);
insert into sales1 values (102,52,789,'30-Mar-2005',545,52312);
insert into sales1 values (103,53,415,'11-Apr-2005',42,7683);
insert into sales1 values (104,45,365,'9-Jun-2005',54,2389);
insert into sales1 values (105,72,457,'2-Jul-2005',23,563912);
insert into sales1 values (106,8,865,'25-Sep-2005',21,635612);
insert into sales1 values (107,87,544,'12-Oct-2005',20,54662);
insert into sales1 values (108,75,341,'5-Dec-2005',45,54623);
```

```
select table_name, partition_name, high_value,num_rows FROM
user_tab_partitions where table_name = 'SALES1';
```

```
select * from sales1 partition(sale_q1);
select * from sales1 partition(sale_q2);
select * from sales1 partition(sale_q3);
select * from sales1 partition(sale_q4);
```

Q2 (R – Basics & Objects)

Write an R program to create a vector, matrix, and data frame. Perform cbind() and rbind() operations and display the structure of each object.

Solution:

```
> #create vector
> my_num = c(2,4,9,0)
> my_num
[1] 2 4 9 0
> class(my_num)
[1] "numeric"
> length(my_num)
[1] 4
> #create matrix
> m <- matrix(c(11,12,13,55,60, 65, 66,72,78), nrow=3, ncol=3)
> m
      [,1] [,2] [,3]
[1,]   11   55   66
[2,]   12   60   72
[3,]   13   65   78
> dim(m)
[1] 3 3
> #By Row
> m <- matrix(c(11,12,13,55,60, 65, 66,72,78), nrow=3, ncol=3, byrow=TRUE)
> m
      [,1] [,2] [,3]
[1,]   11   12   13
[2,]   55   60   65
[3,]   66   72   78
> #create dataframe
> students.data <- data.frame(
+   sid = (1:3),
+   sname = c("Mukund", "Omkar", "Aman")
+   #stringsAsFactors = FALSE
+ )
> print(students.data)
  sid sname
1   1 Mukund
2   2  Omkar
3   3   Aman
> str(students.data)
'data.frame':   3 obs. of  2 variables:
 $ sid : int  1 2 3
 $ sname: chr  "Mukund" "Omkar" "Aman"
> print(summary(students.data))
      sid      sname
Min.   :1.0   Length:3
1st Qu.:1.5   Class :character
Median :2.0   Mode  :character
Mean   :2.0
```

3rd Qu.:2.5

Max. :3.0

```
> dim(students.data)
```

```
[1] 3 2
```

```
> v1 = c(10,20,30,40,50)
```

```
> v2 = c(101,102,103,104,105)
```

```
> z1 = cbind(v1,v2)
```

```
> z1
```

```
      v1  v2
```

```
[1,] 10 101
```

```
[2,] 20 102
```

```
[3,] 30 103
```

```
[4,] 40 104
```

```
[5,] 50 105
```

```
> z2 = rbind(v1,v2)
```

```
> z2
```

```
 [,1] [,2] [,3] [,4] [,5]
```

```
v1   10   20   30   40   50
```

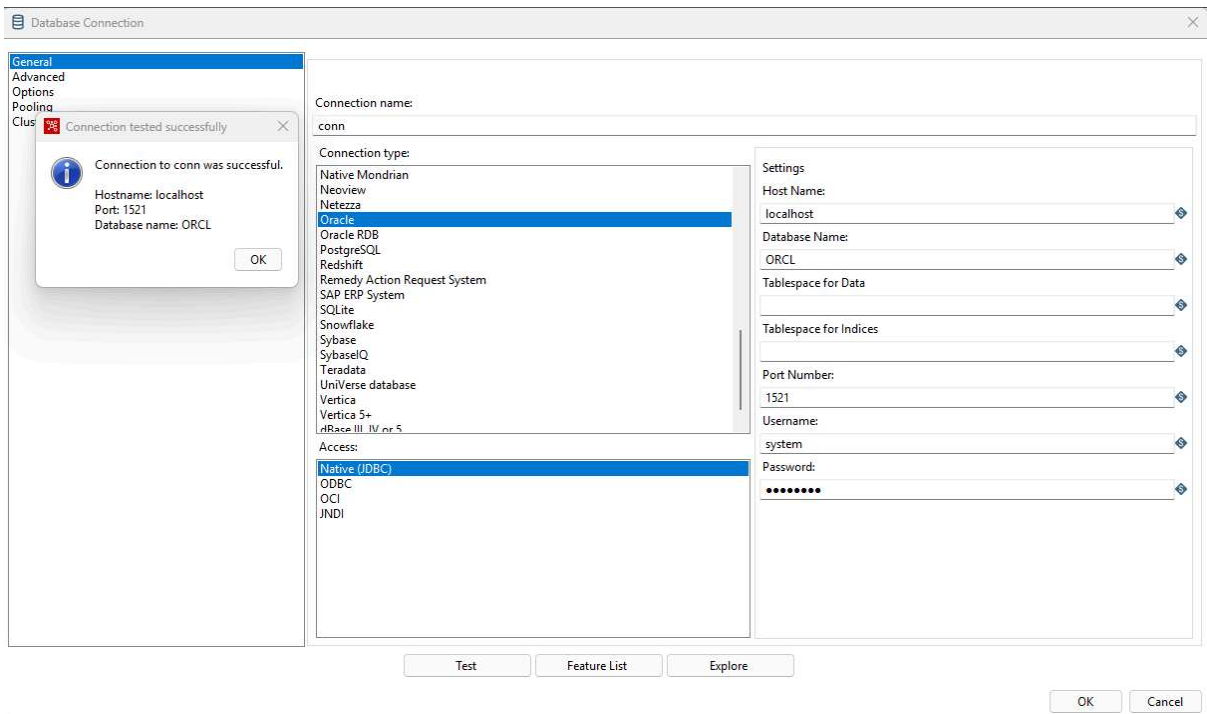
```
v2  101  102  103  104  105
```

Q3 (Pentaho – Table to Table ETL)

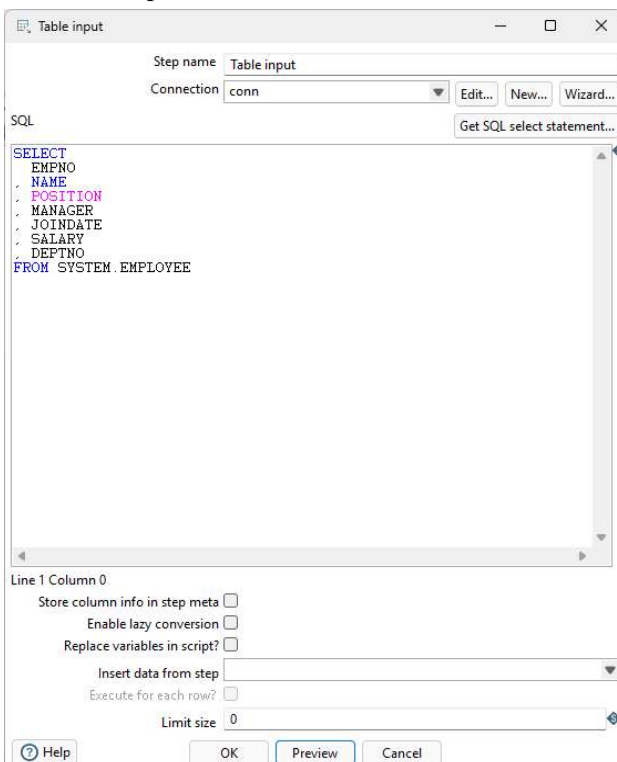
Using Pentaho Data Integration, copy data from a source table to a target table in Oracle.
Explain the steps involved and show the output.

Solution:

Data Connection:



Get Table Input:



Preview Input Table:

Examine preview data

Rows of step: Table input (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103

Close Show Log

Specify Target Table & Get Datafields

Table output

Step name: Table output

Connection: conn [Edit... New... Wizard...]

Target schema: [Browse...]

Target table: output [Browse...]

Commit size: 1000

Truncate table: ☒

Ignore insert errors: ☐

Specify database fields: ☒

Main options Database fields

Fields to insert:

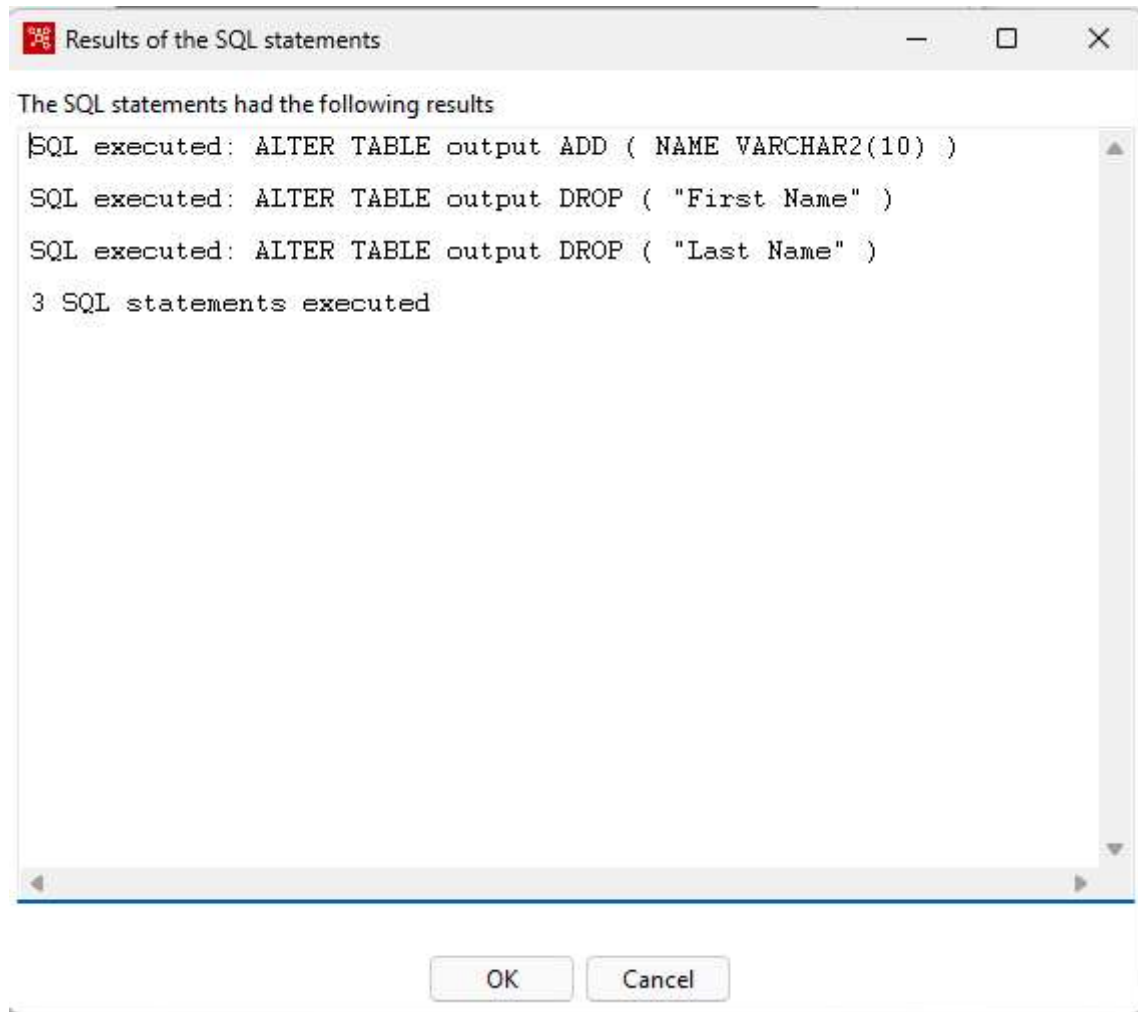
#	Table field	Stream field
1	EMPNO	EMPNO
2	NAME	NAME
3	POSITION	POSITION
4	MANAGER	MANAGER
5	JOINDATE	JOINDATE
6	SALARY	SALARY
7	DEPTNO	DEPTNO

[Get fields]

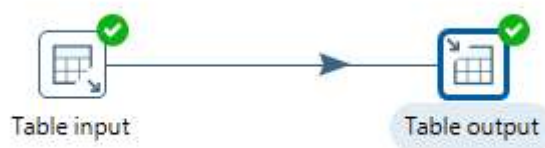
Enter field mapping

[?] Help OK Cancel SQL

And Click on SQL



Design as Below:



Now Preview Output Table

Examine preview data

Rows of step: Table output (14 rows)

#	EMPNO	NAME	POSITION	MANAGER	JOINDATE	SALARY	DEPTNO
1	1	abc	xyz	201	2006/03/15 00:00:00.000000000	40000	101
2	2	e2	p1	201	2016/05/15 00:00:00.000000000	30000	101
3	3	e3	p2	201	2005/06/28 00:00:00.000000000	20000	103
4	4	e4	p1	201	1972/03/05 00:00:00.000000000	15000	102
5	5	e5	p3	201	2016/04/25 00:00:00.000000000	35000	103
6	7	e6	p3	201	2016/04/25 00:00:00.000000000	15000	101
7	8	e7	p2	202	2018/01/12 00:00:00.000000000	28000	102
8	9	e8	p1	202	2015/09/03 00:00:00.000000000	32000	101
9	10	e9	p3	201	2019/11/20 00:00:00.000000000	36000	103
10	11	e10	xyz	203	2014/02/17 00:00:00.000000000	41000	104
11	12	e11	p1	202	2011/05/29 00:00:00.000000000	29000	101
12	13	e12	p2	203	2017/08/08 00:00:00.000000000	22000	102
13	14	e13	p3	202	2013/07/19 00:00:00.000000000	37000	104
14	15	e14	xyz	201	2009/12/06 00:00:00.000000000	43000	103

Close